Thurstone is the Model of Choice

Renormalization fails the link between first-place shares and pairwise margins in eleven datasets

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Abstract

A population ranks or chooses among alternatives, and two things are observed: how often each alternative is placed first, and how often one alternative is placed above a particular rival. Luce's axiom ties these together by renormalization, so that restricting attention to a pair leaves its odds unchanged. Thurstone's model of 1927 treats the shares as winning probabilities in a race, and predicts that a two-item field compresses those odds by a computable amount. We compare the two on ten archival ranking datasets and one experiment covering some 148,000 dataset-responses, and report two findings.

First, the nesting class is empirically necessary. That the contest framework nests renormalization is a theorem, not a finding: by Yellott's result the axiom holds exactly where performance noise is Gumbel. What the data add is that this nested point is rejected everywhere, so the surrounding class is required rather than merely available. Pairwise odds contract by between 0.19 and 0.59 on the log-odds scale where the axiom requires exactly zero.

Second, the untuned default within that class is already better. Unit-variance independent Gaussian noise, Thurstone's Case V, with no quantity fitted to the data being predicted, is the closer prediction in eleven of eleven datasets and gives lower held-out log loss in eight of the eight that admit the test, by 0.002 to 0.011 nats per prediction.

In one line: when studying human choice, Gaussian contest removal is generally superior to renormalization, and it costs nothing extra to compute or to calibrate. What is rejected is renormalization applied to aggregate shares, which is the assumption inside the Plackett–Luce likelihood. Nothing here establishes how individuals choose.

1 Two ways to link first-place shares to pairwise margins

Suppose we know, for a population and a set S of alternatives, the share p_i choosing or first-ranking each item i. Luce's axiom [11] assigns each item a fixed worth $u(i)$ with

$$p_i = \frac{u(i)}{\sum_{j \in S} u(j)}, \tag{1}$$

so the worths are determined up to scale by the shares. The axiom then fixes what happens on any subset: the denominator changes and nothing else does, leaving p_i/p_j untouched for any two survivors. In particular the share preferring i to j when only those two are considered must equal $p_i/(p_i + p_j)$. This is Independence of Irrelevant Alternatives, and in the analysis of rankings it is the content of the Plackett–Luce likelihood.

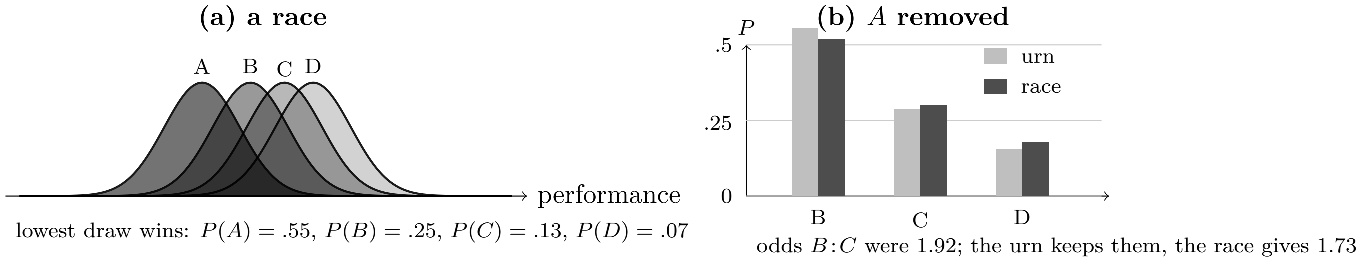

Figure 1: The two contenders disagree only under restriction. In (a) each item draws a performance and the lowest wins. In (b) A is removed. The urn rescales, so every surviving odds ratio is unchanged by construction. The race is re-run without A and the vacated mass is redistributed unevenly, moving the $B\!:\!C$ odds from 1.92 to 1.73. Locations are calibrated to the probabilities in (a) by the lattice method of [4]; the numbers are exact.

Thurstone [18] links the same two quantities through a race. Item i has a location a_i and draws a performance $X_i \sim N(a_i, 1)$ on each occasion, and the item with the winning draw is chosen, so p_i is a winning probability in a field of $|S|$. The locations are recovered from the shares, and the pairwise margin then follows from a two-horse race between i and j alone. It is smaller in magnitude than the odds the shares suggest, because beating one rival is easier than beating several and a favourite's advantage counts for less when the field is short.

Figure 1 shows the two predictions on a four-item example. Both reproduce the four shares exactly, since each has enough freedom to do so. They disagree about the pairwise margin, and they disagree by a definite amount.

This paper makes two claims. They are logically independent, they are supported by different evidence, and conflating them has been the main source of overstatement in this literature, so they are stated separately here and referred to as Claim 1 and Claim 2 throughout.

Claim 1. The nesting class is empirically necessary. Yellott [23] showed that Luce's axiom holds within additive random utility models exactly when the performance noise is Type I extreme value, so renormalization is not a rival of the contest framework but a point inside it, the point at which the noise is Gumbel. Since the data reject that point in every dataset examined, what is required is the surrounding class. The extra flexibility is not a modelling preference; it is forced, and it comes with the axiom nested inside it as a special case that can be recovered whenever the data want it.

Claim 2. The untuned default in that class is already better. Within the contest class one might expect that fitting the noise, or at least choosing it with the data in view, would be necessary before beating the Gumbel point. It is not. Unit-variance independent Gaussian noise, the case Mosteller [16] made tractable and the one every textbook reaches for, is closer to the observed pairwise margins in eleven of eleven datasets and gives lower held-out log loss in eight of the eight that admit the test. Nobody selected it to fit these data, no quantity in it was tuned to them, and it still wins everywhere we look. The improvement is therefore available without any fitting exercise, which is what makes Claim 2 practically useful and also what makes it surprising.

Neither prediction involves a quantity fitted to the pairwise data. Both calibrate $|S| - 1$ relative worths or locations from the shares alone, and then predict. We therefore compare zero-extra-parameter predictions, in the sense that the pairwise observations play no part in the calibration of either model.

Recovering locations from shares under Equation (1)'s competitor requires an integral. With independent equal-variance normals it is one-dimensional for any number of alternatives,

$$p_i = \int \phi(x - a_i) \prod_{k \neq i} \big[1 - \Phi(x - a_k)\big]\, dx, \tag{2}$$

since conditioning on $X_i = x$ makes the other draws independent events. What is genuinely high-dimensional is the correlated case, multinomial probit, and the likelihood of a complete ranking rather than of a top choice. The practical obstacle for most of the twentieth century was that Equation (2) must be evaluated numerically, and inverted numerically, once per candidate location vector inside an optimisation. A lattice method [4] now returns the whole vector of winning probabilities for fields of 10^5, which is what makes the inversion routine here.

2 What is observed, and at what level

Ten of the eleven datasets are complete rankings. From a ranking we can read off which alternative the respondent placed first, and also which of any two alternatives they placed higher. Both quantities are transformations of the same response, so the comparison in those ten datasets is between two accounts of one ranking distribution: the axiom says the pairwise margin is determined by the first-place shares in a particular way, and the race says it is determined differently.

That is not the same as observing choice when a menu is physically restricted. A respondent's ranking imposes a single ordering that applies to every subset by construction, so no ranking dataset can show a person preferring i to j in one menu and j to i in another, and none can reveal whether the person would in fact choose as their ranking implies when handed only two options. Any statement about menu restriction from these ten datasets rests on the assumption that they would. The eleventh dataset makes no such assumption, because subjects there chose separately from every subset.

The level of analysis matters as much as the elicitation. All eleven datasets give population quantities, and the aggregate failure of Independence of Irrelevant Alternatives does not imply that individuals violate it. A population of people who each choose by Equation (1), with worths differing from person to person, generally does not satisfy the axiom in aggregate, which is why mixed logit is a useful model and why its probabilities are integrals over heterogeneous logit probabilities rather than logit probabilities themselves [15]. So three claims must be kept apart. The first is that a single representative-agent Luce model, calibrated to aggregate first-place shares, predicts the aggregate pairwise margin. This paper rejects that claim. The second is that individuals have heterogeneous worths and each choose by a Luce rule, which would be consistent with everything reported here. The third is that a given individual, asked repeatedly, chooses by a Luce rule; nothing here addresses it, and the within-subject evidence in Section 7 speaks only to whether individual choices are order-consistent across menus.

3 Prior evidence

Independence of Irrelevant Alternatives has been contradicted repeatedly, and usually by manipulating the set. Debreu [5] pointed out that adding a close substitute should divide that alternative's support rather than draw proportionally from everything. Tversky [22] showed that similarity defeats simple scalability, of which the axiom is a special case. Huber, Payne and Puto [9] found that a dominated decoy raises the target's absolute share, which no random utility model allows, and Simonson [17] reported a related effect for compromise options. Luce [12] himself treated the empirical case as closed by 1977.

Econometrics kept the logit and relaxed the independence around it, through nested logit, the generalized extreme value family, and mixed logit [15], with a specification test from Hausman and McFadden [21]. The Thurstonian option was written down as multinomial probit [8] and reported as practical for only a few alternatives; simulation made it estimable later and it remains costly [19]. Mixed logit is often described as retaining the logit's convenience, but its choice probabilities are integrals requiring numerical or simulated evaluation.

Two results from this literature shape the present test. Yellott [23] showed that within additive random utility models with independent and identically distributed noise, and with menus rich enough to identify it, the axiom holds exactly when the noise is Type I extreme value, or Gumbel. The phrase double exponential is sometimes used for this law and sometimes for the Laplace; the Gumbel is meant. That result places the two accounts inside one framework, with their disagreement reduced to the shape of the noise, and it is why a single statistic can separate them. Block and Marschak [1] and Falmagne [6] characterised when a complete collection of choice probabilities admits any random utility representation, which Section 7 uses.

What has not been done, so far as we can establish, is to calibrate both accounts to the same aggregate shares and ask which predicts the pairwise margin. The classic demonstrations establish that an effect exists, on two- and three-item menus built from constructed profiles where there is no larger field to calibrate from. The econometric alternatives fit a covariance or a mixing distribution, so their success leaves open how much the unadorned race would have achieved.

4 Data

Eleven datasets have human respondents, complete rankings or ratings fine enough to yield one, and enough respondents for stable shares. Two rank-order card tasks from the General Social Survey ask what a child should learn and what matters in a job. Three independent samples of Jester users rate the same ten jokes on a continuous scale. Sushi rankings come from a Japanese web survey and film rankings are induced from Netflix ratings. There are perceptual discrimination tasks on dot arrays and puzzles, rankings of seven sports by participation preference, rankings of ten occupations by perceived prestige, and rankings of four political goals from a five-nation survey. The eleventh is the experiment of Costa-Gomes and colleagues [3], in which subjects chose from every subset of five consumer products; the archive identifies the products only by two-letter codes, so we describe them no more precisely than the archive does.

Counting is not straightforward and we report it plainly. The eleven rows comprise 147,719 dataset-responses. Unique people are fewer: the two General Social Survey tasks were fielded to overlapping samples of the same survey, while the three Jester files are disjoint user groups.

Two families of source were tried and set aside. Rating scales with eleven points cannot produce rankings of more than about five items, because respondents use only five or six of the levels available to them; in comparative election studies between 0.3 and 5 per cent of respondents give a tie-free ordering of nine parties. Preferential ballots fail for a different reason. Voters rank a few candidates and leave the rest unmarked, which the files record as a tie at the bottom, and reading that tie as indifference gives $\lambda = 0.68$ across fifty-three files where counting only pairs a voter actually ordered gives 0.15. The instrument also fails to replicate: twelve elections of one body, with the same five-candidate format and about ten thousand ballots each, give values spread from 0.36 to 2.56, far wider than sampling error at those sample sizes. Ballots are excluded.

5 Held-out predictive comparison

The comparison that decides the matter scores both accounts as probability forecasts on data neither has seen. Respondents are split into five folds. On the training folds we compute first-place shares over the full item set, and calibrate both accounts to those shares alone: the worths for renormalization are the shares themselves, and the race's locations come from inverting Equation (2). Each account then yields a predicted distribution on every subset of at least two items. For each held-out respondent and each subset, the outcome is that respondent's highest-ranked member of the subset, and we accumulate log loss. Log loss is a proper scoring rule, so neither account can gain by distorting a forecast that is already right, and no pairwise or subset outcome enters either calibration.

Dataset	n	K	Subsets	Renorm.	Race	Gain
Sushi	5,000	10	1,013	1.3727	1.3614	+0.0113 [+0.0096, +0.0130]
Political goals	2,262	4	11	0.8836	0.8767	+0.0069 [+0.0060, +0.0079]
GSS socialization	5,000	5	26	0.7986	0.7929	+0.0057 [+0.0043, +0.0072]
Sports participation	130	7	120	1.2594	1.2538	+0.0056 [+0.0021, +0.0094]
Jester file 3	5,000	10	1,013	1.5189	1.5165	+0.0024 [+0.0016, +0.0032]
Jester file 1	5,000	10	1,013	1.5287	1.5267	+0.0020 [+0.0013, +0.0027]
Jester file 2	5,000	10	1,013	1.5247	1.5230	+0.0018 [+0.0010, +0.0026]
GSS job values	5,000	5	26	0.8593	0.8575	+0.0018 [+0.0005, +0.0031]

Table 1: Mean held-out log loss per prediction, over every subset of size two or more, with both accounts calibrated only to training-fold first-place shares. Gain is the loss under renormalization minus the loss under the race, so positive favours the race; intervals are respondent bootstraps. The race is better in eight of eight datasets.

The race predicts held-out restricted choices better in every dataset, and every interval excludes zero. The margins are small in absolute terms, between 0.002 and 0.011 nats per prediction against losses near one nat, so the honest summary is that the race is reliably better rather than dramatically better. Datasets are capped at five thousand respondents for runtime, and the occupational prestige rankings are omitted here because some alternatives are never ranked first in a training fold, leaving renormalization with a zero worth and no prediction to score.

5.1 The contraction that drives it

The direction of the difference is easier to see in a single summary statistic than in a log score. Let p be the first-place shares and q_{ij} the share placing i above j. The shift

$$\delta_{ij} = \log \frac{q_{ij}}{1 - q_{ij}} - \log \frac{p_i}{p_j} \tag{3}$$

is zero for every pair under renormalization, and negative under the race. Fitting

$$\hat{\lambda} = \frac{\sum_{i<j} \log(p_i/p_j) \left(-\delta_{ij}\right)}{\sum_{i<j} \left[\log(p_i/p_j)\right]^2} \tag{4}$$

through the origin over unordered pairs, weighting each equally and dropping pairs where a log odds is undefined, gives the values in Table 2.

Dataset	Domain	Responses	$\hat{\lambda}$	Race predicts	Race error
GSS job values	values	21,797	0.227	0.234	0.007
Dots and puzzles	perception	6,363	0.188	0.161	0.027
Jester file 3	humour	24,904	0.374	0.337	0.037
GSS socialization	values	33,921	0.276	0.314	0.038
Jester file 1	humour	24,953	0.393	0.350	0.043
Sports participation	leisure	130	0.347	0.304	0.043
Jester file 2	humour	23,494	0.391	0.343	0.048
Sushi and Netflix	food, film	9,379	0.420	0.317	0.103
Consumer products	menu choice	373	0.419	0.231	0.188
Occupational prestige	social	143	0.591	0.300	0.291
Political goals	values	2,262	0.425	0.181	0.244

Table 2: Observed contraction and the race's prediction. Renormalization predicts zero, so its error equals the observed value in every row, from 0.188 to 0.591. Responses are dataset-responses; Section 4 discusses unique respondents.

Renormalization is wrong by the whole of its prediction everywhere. The race's error is under 0.05 in seven rows. Its residual has both signs: it under-predicts contraction in nine datasets and over-predicts in the two General Social Survey tasks, where observed values of 0.227 and 0.276 fall below predictions of 0.234 and 0.314. The three largest under-predictions come from the datasets with the fewest usable pairs, six, ten and fourteen, so we do not read a domain pattern from them.

The eleven rows are not eleven independent experiments. Three are Jester files, two are General Social Survey items, and several arrive through the same repositories, so the count describes the table rather than supporting a claim about independent replications. The held-out comparison in Table 1 is the basis for inference here, and its intervals are respondent bootstraps within dataset.

6 What kind of noise would fit better

Since the race under-predicts contraction more often than not, it is worth asking which noise laws contract more than a Gaussian, and the question has a clean answer that runs against intuition. Simulating populations whose members are races with a common noise law, standardised to unit variance, and computing the same λ:

Heavier tails move the model toward the axiom rather than away from it, so a heavy-tailed race is the wrong candidate for datasets that contract more than Gaussian. Bounded or thinner-tailed noise is the right direction there. Skewness also reduces contraction, which makes it a candidate for the two General Social Survey rows, where observed contraction falls short of the Gaussian prediction. Other explanations are not excluded by this exercise and are not

Noise law	λ
Gumbel	0.001
Student t, 3 degrees of freedom	0.121
Student t, 8 degrees of freedom	0.203
Skew-normal, $a = 4$	0.078
Gaussian	0.245
Uniform on a bounded interval	0.352

Table 3: Contraction produced by different performance-noise laws in simulated races. The Gumbel row recovers the axiom, as Yellott's theorem requires, and also checks the pipeline. Heavier tails contract less than the Gaussian; bounded support contracts more.

distinguished by it, among them unequal variances across alternatives, correlated performances, heterogeneity across respondents, and utilities that depend on the set.

7 Choices from every subset

The consumer-products experiment observes what the ranking datasets cannot. Its 373 subjects chose from every subset of five goods, one subject at a time, so a reversal is observable rather than excluded by construction, and the choices are individual rather than aggregate.

Reversals are uncommon. Of 26,099 comparisons in which an item chosen from a larger menu was still available in a smaller one, 829 switched away from it, a rate of 3.2 per cent; among subjects required to choose rather than permitted to defer, and so free of any coding decision, the rate is 3.9 per cent. The subject-level distribution is uneven. Of 367 subjects, 235 never reverse, the median subject reverses never, the ninetieth percentile reverses eight times, and the most violating tenth of subjects account for 58 per cent of all reversals. So most individuals are order-consistent across menus and a minority are markedly not, which is a pattern an average rate conceals.

Pooling subjects gives choice probabilities for all thirty-one menus, and the Block–Marschak sums test whether those probabilities admit any random utility representation. Of eighty computable sums, nine fall below zero and one below -0.02, the worst at -0.055; among forced-choice subjects sixteen fall below zero and ten below -0.02, the worst at -0.088. Estimated probabilities can produce small negative sums by sampling noise alone, and we have not computed the distance from the random utility polytope or intervals for the individual sums, so we report the counts and draw no strong conclusion. The complete-ranking datasets cannot contribute to this question at all, since a distribution over rankings is a random utility representation by construction.

8 Discussion

Claim 1, restated. The nesting class is empirically necessary. The nesting is Yellott's theorem; what the data establish is that the nested point cannot be retained. Renormalization occupies one point in that class, the point where performance noise is Gumbel, and every dataset here rejects it: the pairwise margin contracts relative to the first-place shares by between 0.19 and 0.59 on the log-odds scale, where the axiom requires exactly zero. Because the axiom is nested rather than adjacent, adopting the class costs nothing in generality. It can reproduce renormalization whenever a population turns out to behave that way, and the simulations in Section 6 confirm that it does so exactly, returning a contraction of 0.001 when the noise is Gumbel.

Claim 2, restated. The untuned Gaussian default is closer than renormalization in every dataset examined. Unit-variance independent Gaussian noise was chosen a century ago for tractability rather than fit, and nothing in it was adjusted to these datasets, yet it is the closer prediction in eleven of eleven and the better forecast in eight of eight under a proper score. The margins on the log score are small, between 0.002 and 0.011 nats per prediction, so the practical gain is modest even though its direction is consistent. What makes this worth reporting is the absence of tuning: a practitioner who reaches for Case V rather than renormalization, with no data in hand beyond aggregate shares, is better off in every case examined here.

The boundaries of both claims should be stated as plainly as their content. What is rejected is a representative-agent Luce model of a population, which is the assumption inside the Plackett–Luce likelihood used throughout ranking analysis. Heterogeneous individuals who each choose by a Luce rule would generate aggregate violations of this kind, so nothing here shows that individual choice departs from logit, and the within-subject experiment speaks to order-consistency rather than to functional form. Ten of the eleven datasets compare two accounts of a ranking distribution rather than observing menu restriction, and their contribution should be read as such.

Case V is also not the end of the matter. Its residual has both signs, and Section 6 identifies the direction each sign points: bounded or thinner-tailed noise where contraction exceeds the Gaussian prediction, skewness or heavier tails where it falls short. Whether one noise law fits all eleven datasets is answerable with the same data and is not answered here.

Three limitations bound what has been shown. The political goals, occupational prestige and consumer datasets rest on six, ten and fourteen usable pairs, so their contraction estimates are the least secure in Table 2. The held-out comparison caps datasets at five thousand respondents for runtime and omits occupational prestige, where an alternative is never ranked first in some training fold and renormalization has no prediction to offer. And nothing here addresses regularity violations of the decoy and compromise kind, which lie outside both accounts.

Both claims reduce to one practical statement. When the object of study is human choice and what is available is a set of aggregate shares, Gaussian contest removal is generally superior to renormalization, and it requires no extra parameter, no extra data and no fitting step.

For psychometrics the consequence concerns defaults. Plackett–Luce and Bradley–Terry likelihoods are standard for rankings and paired comparisons and they carry the axiom with them, largely because the contest was once impractical to compute. That reason has expired, the axiom is available as a nested special case, and on this evidence the ordering of defaults should be reversed for population-level ranking data.

Data and code

Sushi rankings come from Kamishima [10]; Jester ratings from the Eigentaste project [7], gauge-set files one to three; the General Social Survey cumulative file from NORC; the perceptual, leisure, prestige and political-goal rankings via PrefLib [13] and the CRAN packages BayesMallows, ConsRank, PerMallows and PLMIX; the menu-choice data from the replication archive of [3]. All are public and free. Analysis code, derived tables and cached copies are under `research/human` in `github.com/microprediction/winning`, together with the lattice calibration.

Two storage conventions must be told apart when reproducing this. BayesMallows, ConsRank and PerMallows store ranks, so element j holds the rank of alternative j, while PLMIX stores orderings, so the first element names the favourite. Reading one as the other transposes the data silently, and no error is raised. We established which is which by cross-validation: the political goals data ships in two of these packages, and the two agree only when each is read under its own convention.

References

[1] H. D. Block and Jacob Marschak. Random orderings and stochastic theories of responses. In *Contributions to Probability and Statistics*, Stanford University Press, 1960.

[2] Ralph A. Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. The method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.

[3] Miguel A. Costa-Gomes, Carlos Cueva, Georgios Gerasimou, and Matúš Tejivščák. Choice, deferral and consistency. *Quantitative Economics*, 13:1297–1318, 2022.

[4] Peter Cotton. Inferring relative ability from winning probability in multientrant contests. *SIAM Journal on Financial Mathematics*, 12(1), 2021.

[5] Gerard Debreu. Review of R. D. Luce, *Individual Choice Behavior*. *American Economic Review*, 50:186–188, 1960.

[6] Jean-Claude Falmagne. A representation theorem for finite random scale systems. *Journal of Mathematical Psychology*, 18:52–72, 1978.

[7] Ken Goldberg, Theresa Roeder, Dhruv Gupta, and Chris Perkins. Eigentaste: a constant time collaborative filtering algorithm. *Information Retrieval*, 4(2):133–151, 2001.

[8] Jerry A. Hausman and David A. Wise. A conditional probit model for qualitative choice. *Econometrica*, 46(2):403–426, 1978.

[9] Joel Huber, John W. Payne, and Christopher Puto. Adding asymmetrically dominated alternatives: violations of regularity and the similarity hypothesis. *Journal of Consumer Research*, 9:90–98, 1982.

[10] Toshihiro Kamishima. Nantonac collaborative filtering: recommendation based on order responses. In *Proceedings of KDD*, 2003.

[11] R. Duncan Luce. *Individual Choice Behavior: A Theoretical Analysis*. Wiley, 1959.

[12] R. Duncan Luce. The choice axiom after twenty years. *Journal of Mathematical Psychology*, 15(3):215–233, 1977.

[13] Nicholas Mattei and Toby Walsh. PrefLib: a library of preference data. In *Algorithmic Decision Theory*, 2013.

[14] Daniel McFadden. Conditional logit analysis of qualitative choice behavior. In *Frontiers in Econometrics*, Academic Press, 1974.

[15] Daniel McFadden and Kenneth Train. Mixed MNL models for discrete response. *Journal of Applied Econometrics*, 15(5):447–470, 2000.

[16] Frederick Mosteller. Remarks on the method of paired comparisons. *Psychometrika*, 16(1):3–9, 1951.

[17] Itamar Simonson. Choice based on reasons: the case of attraction and compromise effects. *Journal of Consumer Research*, 16:158–174, 1989.

[18] Louis L. Thurstone. A law of comparative judgment. *Psychological Review*, 34:273–286, 1927.

[19] Kenneth E. Train. *Discrete Choice Methods with Simulation*, second edition. Cambridge University Press, 2009.

[20] Amos Tversky and Itamar Simonson. Context-dependent preferences. *Management Science*, 39(10):1179–1189, 1993.

[21] Jerry Hausman and Daniel McFadden. Specification tests for the multinomial logit model. *Econometrica*, 52(5):1219–1240, 1984.

[22] Amos Tversky. Elimination by aspects: a theory of choice. *Psychological Review*, 79(4):281–299, 1972.

[23] John I. Yellott. The relationship between Luce's choice axiom, Thurstone's theory of comparative judgment, and the double exponential distribution. *Journal of Mathematical Psychology*, 15:109–144, 1977.